Trade-off between plant and soil carbon storage under elevated CO_2

C. Terrer[1,2*], R.P. Phillips[3], B.A. Hungate[4,5], J. Rosende[6], J. Pett-Ridge[1], M. Craig[3,7], K.J. van Groenigen[8], T.F. Keenan[9,10], B.N. Sulman[7], B.D. Stocker[11,12], P.B. Reich[13,14], A.F.A. Pellegrini[2,15], E. Pendall[14], H. Zhang[16], R.D. Evans[17], Y. Carrillo[14], J. B. Fisher[18,19], K. Van Sundert[20], Sara Vicca[20], R.B. Jackson[2].

Affiliations

[1]Physical and Life Sciences Directorate, Lawrence Livermore National Laboratory, Livermore, CA, 94550 USA.
[2]Department of Earth System Science, Stanford University, Stanford, CA, USA.
[3]Department of Biology, Indiana University, Bloomington, IN, 47405, USA.
[4]Center for Ecosystem Science and Society, Northern Arizona University, Flagstaff, Arizona 86011, USA.
[5]Department of Biological Sciences, Northern Arizona University, Flagstaff, AZ, USA.
[6]Institut de Ciència i Tecnologia Ambientals, Universitat Autònoma de Barcelona, Barcelona, Spain.
[7]Environmental Sciences Division and Climate Change Science Institute, Oak Ridge National Laboratory, Oak Ridge, TN, USA.
[8]Department of Geography, College of Life and Environmental Sciences, University of Exeter, Exeter, EX4 4 RJ, UK.
[9]Department of Environmental Science, Policy and Management, UC Berkeley, Berkeley, CA, USA.
[10]Climate and Ecosystem Sciences Division, Lawrence Berkeley National Laboratory, Berkeley, CA, USA.
[11]Department of Environmental Systems Science, ETH, Universitätsstrasse 2, 8092 Zürich, Switzerland
[12]Swiss Federal Institute for Forest, Snow and Landscape Research WSL, Birmensdorf, Switzerland
[13]Department of Forest Resources, University of Minnesota, St. Paul, MN, USA.
[14]Hawkesbury Institute for the Environment, Western Sydney University, Penrith 2571 Australia.
[15]Department of Plant Sciences, University of Cambridge, Cambridge, United Kingdom
[16]Environmental Change Institute, School of Geography and the Environment, University of Oxford, OX1 3QY, UK
[17]School of Biological Sciences and the Stable Isotope Core Laboratory, Washington State University, Pullman, WA 99164-4236, USA.
[18]Jet Propulsion Laboratory, California Institute of Technology, 4800 Oak Grove Dr., Pasadena, CA, 91109, USA
[19]Joint Institute for Regional Earth System Science and Engineering, University of California at Los Angeles, Los Angeles, CA, 90095, USA
[20]Plants and Ecosystems (PLECO), Biology Department, University of Antwerp, Wilrijk, Belgium

*Correspondence to: terrer@stanford.edu

Abstract

Terrestrial ecosystems remove about 30% of the carbon dioxide (CO_2) emitted by human activities each year[1], yet the persistence of this carbon sink depends partly on how plant biomass and soil organic carbon (SOC) stocks respond to future increases in atmospheric CO_2 (refs. 2,3). Although plant biomass often increases in elevated CO_2 (eCO_2) experiments[4–6], SOC has been observed to increase, remain unchanged or even decline[7]. The mechanisms that drive this variation across experiments remain poorly understood, creating uncertainty in climate projections[8,9]. Here we synthesized data from 108 eCO_2 experiments and found that the effect of eCO_2 on SOC stocks is best explained by a negative relationship with plant biomass: when plant biomass is strongly stimulated by eCO_2, SOC storage declines; conversely, when biomass is weakly stimulated, SOC storage increases. This trade-off appears to be related to plant nutrient acquisition, in which plants increase their biomass by mining the soil for nutrients, which decreases SOC storage. We found that, overall, SOC stocks increase with eCO_2 in grasslands (8±2%) but not in forests (0±2%), even though plant biomass in grasslands increase less (9±3%) than in forests (23±2%). Ecosystem models do not reproduce this trade-off, which implies that projections of SOC may need to be revised.